

Risk Factors for Radiation Maculopathy and Papillopathy after Intraocular Irradiation

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Purpose: To evaluate rates of occurrence and risk factors for radiation maculopathy and radiation papillopathy in patients with choroidal melanoma at high risk for these complications.

Design: Cohort study.

Participants: A total of 558 patients treated with proton irradiation for choroidal melanoma between 1986 and 1996 with small to moderate sized tumors (less than 5 mm in height and 15 mm in diameter) located within 4 disc diameters of the macula or optic nerve and with a median ocular follow-up of 4 years.

Methods: Annual and cumulative rates of each endpoint were estimated using life table approaches. Prognostic factors were evaluated using the Cox proportional hazards regression.

Main Outcome Measures: Radiation maculopathy, radiation papillopathy, and vision loss to worse than 20/100.

Results: Cumulative 5-year rates for radiation maculopathy, radiation papillopathy, and vision loss were 64%, 35%, and 68%, respectively. Complication rates rose as a function of radiation exposure to the macula (P for trend = 0.04) or optic disc (P for trend < 0.001), although dose-response patterns were nonlinear. History of diabetes was a significant risk factor for maculopathy (P < 0.001) and optic neuropathy (P = 0.009).

Conclusions: The onset of radiation vasculopathy is determined primarily by the degree of irradiation exposure to the macula and optic disc. Risk may be enhanced among those with underlying vascular disorders. *Ophthalmology* 1999;106:1571-1578

Radiation maculopathy and papillopathy are common sequelae of therapeutic intraocular irradiation¹ and are the major contributors to vision loss following posterior pole irradiation.² Manifestations of these complications include capillary closure, telangiectasia, microaneurysm formation, hemorrhages, exudates, macular edema, vascular sheathing, disc pallor, disc edema, and nerve fiber layer infarcts. In a previous analysis of patients treated with protons for choroidal melanoma,³ we demonstrated that rates of these complications increase with the irradiation dose delivered to the macula. Although damage is limited to the irradiation field,⁴ most patients are at risk, given the predilection of these tumors to arise in the macular region (Judge H, Li W, Egan

K, Gragoudas E. *Invest Ophthalmol Vis Sci* 1998;39(suppl):600)

To better quantify the risks of these complications and factors associated with their onset, we analyzed data collected prospectively on a large series of patients irradiated for choroidal melanoma. In particular, we were interested in determining the shape of the irradiation dose-response curve and whether any clinical or lifestyle factors influence risk, independent of radiation exposures to the optic disc and macula.

Subjects and Methods

Patients were first treated with proton irradiation at the Harvard cyclotron in 1975.⁵⁻⁷ For the present analysis, we identified the subgroup of patients at high risk for radiation vasculopathy, e.g., with tumors no more than 4 disc diameters from either the optic disc (for analyses of papillopathy) or macula (for analyses of maculopathy), who would otherwise have a good prognosis for useful vision, e.g., having small or moderately sized tumors (less than 5 mm in height and 15 mm in diameter). The analysis was further restricted to United States or Canadian residents treated at the Massachusetts Eye and Ear Infirmary (MEEI) beginning in 1986, when results of baseline and follow-up examinations were first recorded onto standardized forms. There were a total of 577 patients treated from January 1986 through December 1996 meeting all eligibility criteria.

Patients were treated with a total dose of 70 cobalt Gy equivalents (CGEs) delivered in five equal treatments over 7 to 10 days (63.6 proton Gy $\times$ 1.1 relative biologic effectiveness equals 70

Originally received: November 10, 1998.

Revision accepted: May 5, 1999.

Manuscript no. 98540.

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Presented in part at the American Academy of Ophthalmology annual meeting, New Orleans, Louisiana, November 1998.

Supported by the Melanoma Research Fund and by the Retina Research Foundation, Houston, Texas.

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